

Topic D1+D2 HL Paper 2

Study Guide *[106 marks]*

A planet is in a circular orbit around a star. The speed of the planet is constant.
The following data are given:

Mass of planet = 8.0×10^{24} kg

Mass of star = 3.2×10^{30} kg

Distance from the star to the planet $R = 4.4 \times 10^{10}$ m.

1a. Explain why a centripetal force is needed for the planet to be in a circular orbit. *[2 marks]*

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1b. Calculate the value of the centripetal force. *[1 mark]*

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A spacecraft is to be launched from the surface of the planet to escape from the star system. The radius of the planet is 9.1×10^3 km.

- 1c. Show that the gravitational potential due to the planet and the star at the surface of the planet is about $-5 \times 10^9 \text{ J kg}^{-1}$. *[3 marks]*

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- 1d. Estimate the escape speed of the spacecraft from the planet-star system. *[2 marks]*

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The table gives data for Jupiter and three of its moons, including the radius r of each object.

Object	Mass/kg	r / m	Orbital radius around Jupiter/m
Jupiter	1.9×10^{27}	7.1×10^7	
Io	8.9×10^{22}	1.8×10^6	4.9×10^8
Ganymede	1.5×10^{23}	2.6×10^6	1.06×10^9
Callisto	1.1×10^{23}	2.4×10^6	1.88×10^9

2a. Calculate, for the surface of Io, the gravitational field strength g_{Io} due to [2 marks]
the mass of Io. State an appropriate unit for your answer.

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A spacecraft is to be sent from Io to infinity.

2b. Show that the $\frac{\text{gravitational potential due to Jupiter at the orbit of Io}}{\text{gravitational potential due to Io at the surface of Io}}$ is about 80. [2 marks]

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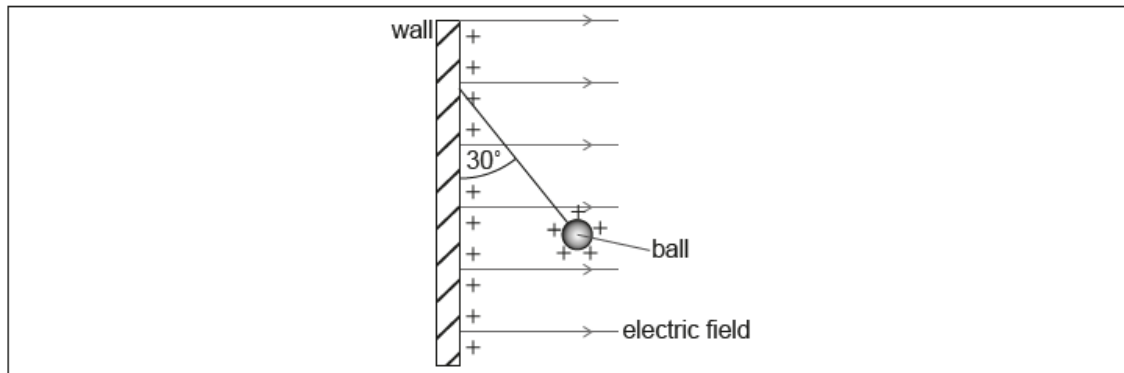
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2c. Outline, using (b)(i), why it is not correct to use the equation [1 mark]

$\sqrt{\frac{2G \times \text{mass of Io}}{\text{radius of Io}}}$ to calculate the speed required for the spacecraft to reach infinity from the surface of Io.

2d. An engineer needs to move a space probe of mass 3600 kg from [2 marks]
Ganymede to Callisto. Calculate the energy required to move the probe from the orbital radius of Ganymede to the orbital radius of Callisto. Ignore the mass of the moons in your calculation.

A vertical wall carries a uniform positive charge on its surface. This produces a uniform horizontal electric field perpendicular to the wall. A small, positively-charged ball is suspended in equilibrium from the vertical wall by a thread of negligible mass.



- 3a. The charge per unit area on the surface of the wall is σ . It can be shown [2 marks] that the electric field strength E due to the charge on the wall is given by the equation

$$E = \frac{\sigma}{2\epsilon_0}.$$

Demonstrate that the units of the quantities in this equation are consistent.

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3b. The thread makes an angle of 30° with the vertical wall. The ball has a mass of 0.025 kg. [3 marks]

Determine the horizontal force that acts on the ball.

[illegible]

3c. The charge on the ball is $1.2 \times 10^{-6} \text{ C}$. Determine σ . [2 marks]

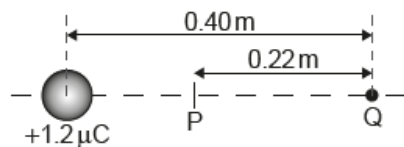
[illegible]

3d. The thread breaks. Explain the initial subsequent motion of the ball. [3 marks]

[illegible]

The centre of the ball, still carrying a charge of $1.2 \times 10^{-6} \text{ C}$, is now placed 0.40 m from a point charge Q . The charge on the ball acts as a point charge at the centre of the ball.

P is the point on the line joining the charges where the electric field strength is zero. The distance PQ is 0.22 m.



3e. Calculate the charge on Q. State your answer to an appropriate number [3 marks] of significant figures.

3f. Outline, without calculation, whether or not the electric potential at P is zero. [2 marks]

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The moon Phobos moves around the planet Mars in a circular orbit.

4a. Outline the origin of the force that acts on Phobos. [1 mark]

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4b. Outline why this force does no work on Phobos. [1 mark]

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4c. The orbital period T of a moon orbiting a planet of mass M is given by [3 marks]

$$\frac{R^3}{T^2} = kM$$

where R is the average distance between the centre of the planet and the centre of the moon.

Show that $k = \frac{G}{4\pi^2}$

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[2 marks]

Mass of Earth = 5.97×10^{24} kg

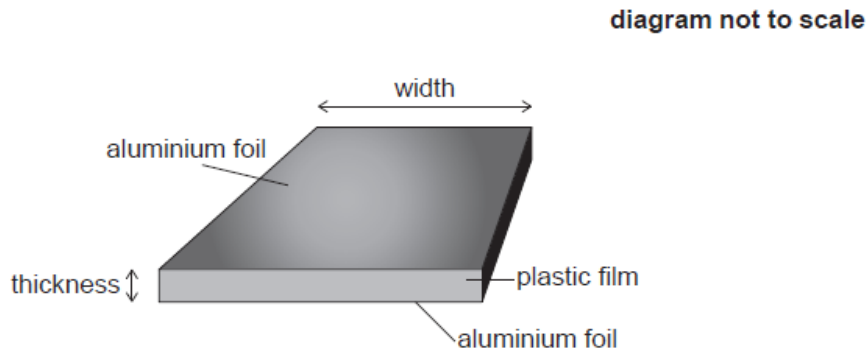
The Earth-Moon distance is 41 times the Mars-Phobos distance.

The orbital period of the Moon is 86 times the orbital period of Phobos.

Calculate, in kg, the mass of Mars.

[illegible]

A student makes a parallel-plate capacitor of capacitance 68 nF from aluminium foil and plastic film by inserting one sheet of plastic film between two sheets of aluminium foil.



The aluminium foil and the plastic film are 450 mm wide.

The plastic film has a thickness of 55 μm and a permittivity of $2.5 \times 10^{-11} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

5a. Calculate the total length of aluminium foil that the student will require. [3 marks]

[illegible]

5b. The plastic film begins to conduct when the electric field strength in it exceeds 1.5 MN C^{-1} . Calculate the maximum charge that can be stored on the capacitor. [2 marks]

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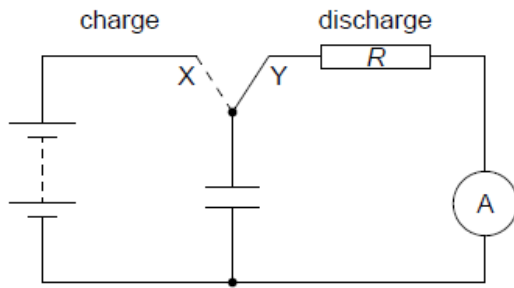
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The student uses a switch to charge and discharge the capacitor using the circuit shown. The ammeter is ideal.



The emf of the battery is 12 V.

- 5c. The resistor R in the circuit has a resistance of $1.2 \text{ k}\Omega$. Calculate the time [3 marks] taken for the charge on the capacitor to fall to 50 % of its fully charged value.

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5d. The ammeter is replaced by a coil. Explain why there will be an induced *[2 marks]* emf in the coil while the capacitor is discharging.

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5e. Suggest **one** change to the discharge circuit, apart from changes to the *[2 marks]* coil, that will increase the maximum induced emf in the coil.

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A planet of mass m is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is V .

6a. Show that the total energy of the planet is given by the equation shown. *[2 marks]*

$$E = \frac{1}{2}mV$$

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6b. Suppose the star could contract to half its original radius without any loss of mass. Discuss the effect, if any, this has on the total energy of the planet. *[2 marks]*

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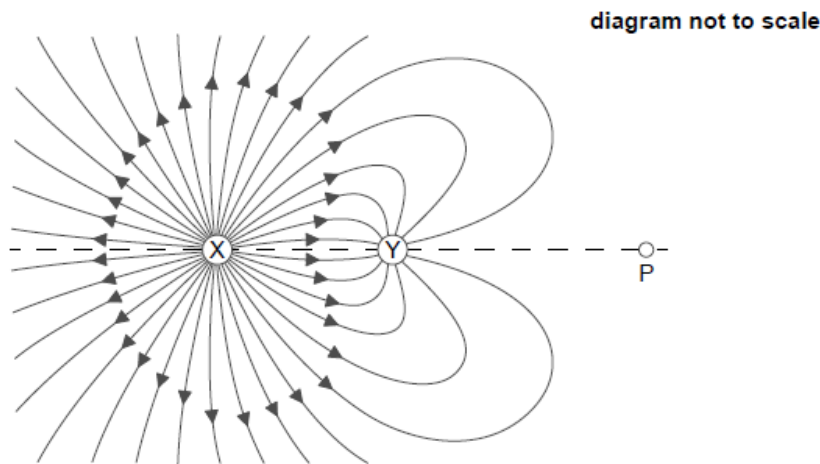
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6c. The diagram shows some of the electric field lines for two fixed, charged [2 marks] particles X and Y.



The magnitude of the charge on X is Q and that on Y is q . The distance between X and Y is 0.600 m. The distance between P and Y is 0.820 m.

At P the electric field is zero. Determine, to **one** significant figure, the ratio $\frac{Q}{q}$.

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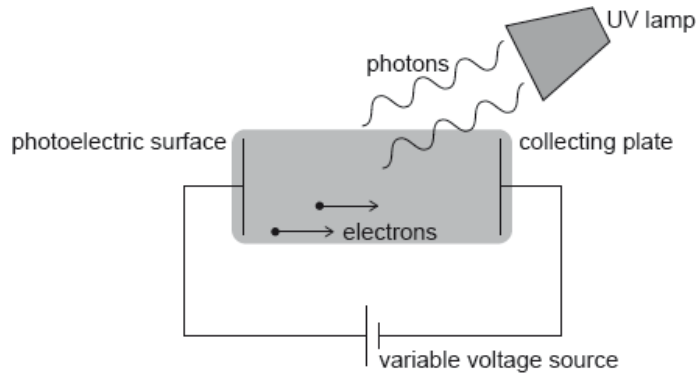
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Hydrogen atoms in an ultraviolet (UV) lamp make transitions from the first excited state to the ground state. Photons are emitted and are incident on a photoelectric surface as shown.



7a. Show that the energy of photons from the UV lamp is about 10 eV. [2 marks]

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The photons cause the emission of electrons from the photoelectric surface. The work function of the photoelectric surface is 5.1 eV.

7b. Calculate, in J, the maximum kinetic energy of the emitted electrons. [2 marks]

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7c. Suggest, with reference to conservation of energy, how the variable voltage source can be used to stop all emitted electrons from reaching the collecting plate. [2 marks]

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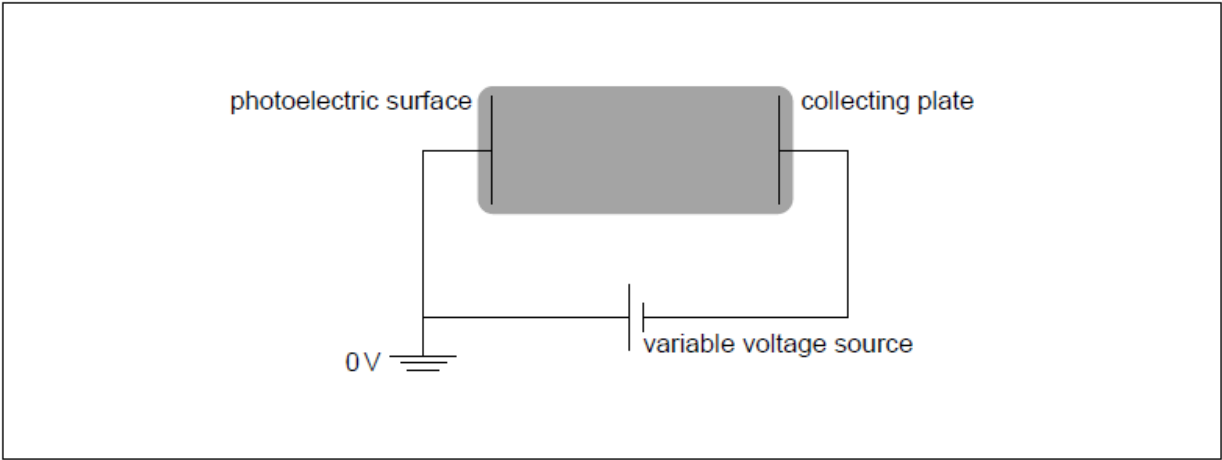
7d. The variable voltage can be adjusted so that no electrons reach the collecting plate. Write down the minimum value of the voltage for which no electrons reach the collecting plate. [1 mark]

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The electric potential of the photoelectric surface is 0 V. The variable voltage is adjusted so that the collecting plate is at -1.2 V.



7e. On the diagram, draw and label the equipotential lines at -0.4 V and -0.8 V. [2 marks]

7f. An electron is emitted from the photoelectric surface with kinetic energy [2 marks]
2.1 eV. Calculate the speed of the electron at the collecting plate.

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A planet has radius R . At a distance h above the surface of the planet the gravitational field strength is g and the gravitational potential is V .

8a. State what is meant by gravitational field strength. [1 mark]

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8b. Show that $V = -g(R + h)$. [2 marks]

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8c. Draw a graph, on the axes, to show the variation of the gravitational potential V of the planet with height h above the surface of the planet. [2 marks]



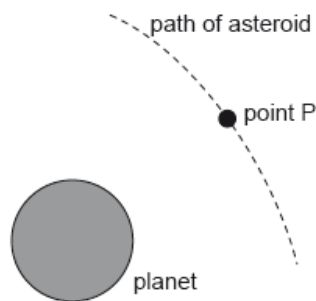
8d. A planet has a radius of 3.1×10^6 m. At a point P a distance 2.4×10^7 m above the surface of the planet the gravitational field strength is 2.2 N kg^{-1} . Calculate the gravitational potential at point P, include an appropriate unit for your answer. [1 mark]

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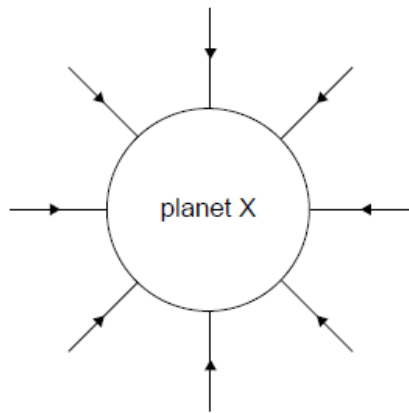
8e. The diagram shows the path of an asteroid as it moves past the planet. [3 marks]



When the asteroid was far away from the planet it had negligible speed. Estimate the speed of the asteroid at point P as defined in (b).

8f. The mass of the asteroid is 6.2×10^{12} kg. Calculate the gravitational force experienced by the **planet** when the asteroid is at point P. [2 marks]

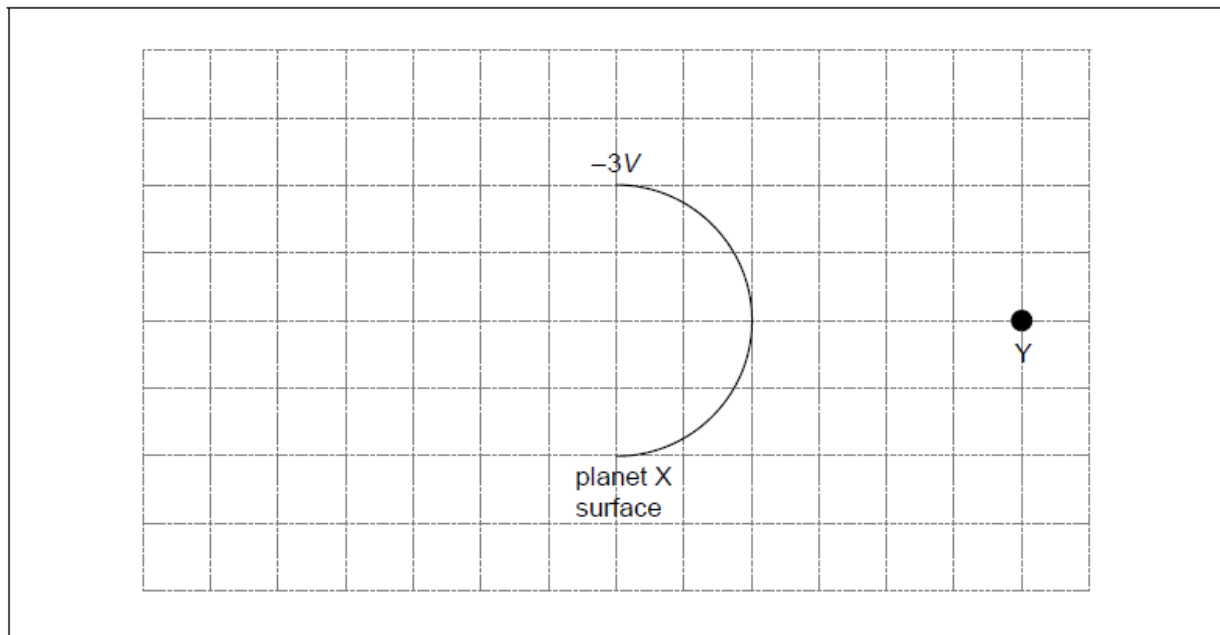
The diagram shows the gravitational field lines of planet X.



- 9a. Outline how this diagram shows that the gravitational field strength of planet X decreases with distance from the surface. [1 mark]

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- 9b. The diagram shows part of the surface of planet X. The gravitational potential at the surface of planet X is $-3V$ and the gravitational potential at point Y is $-V$. [2 marks]



Sketch on the grid the equipotential surface corresponding to a gravitational potential of $-2V$.

9c. A meteorite, very far from planet X begins to fall to the surface with a *[3 marks]*

negligibly small initial speed. The mass of planet X is 3.1×10^{21} kg and its radius is 1.2×10^6 m. The planet has no atmosphere. Calculate the speed at which the meteorite will hit the surface.

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9d. At the instant of impact the meteorite which is made of ice has a *[2 marks]*
temperature of 0 °C. Assume that all the kinetic energy at impact gets
transferred into internal energy in the meteorite. Calculate the percentage of the
meteorite’s mass that melts. The specific latent heat of fusion of ice is 3.3×10^5 J
kg⁻¹.

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The gravitational potential due to the Sun at its surface is $-1.9 \times 10^{11} \text{ J kg}^{-1}$. The following data are available.

Mass of Earth	$= 6.0 \times 10^{24} \text{ kg}$
Distance from Earth to Sun	$= 1.5 \times 10^{11} \text{ m}$
Radius of Sun	$= 7.0 \times 10^8 \text{ m}$

10a. Outline why the gravitational potential is negative.

[2 marks]

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10b. The gravitational potential due to the Sun at a distance r from its centre is V_S . Show that

[1 mark]

$rV_S = \text{constant}$.

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10c. Calculate the gravitational potential energy of the Earth in its orbit around the Sun. Give your answer to an appropriate number of significant figures. [2 marks]

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10d. Calculate the total energy of the Earth in its orbit. [2 marks]

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10e. An asteroid strikes the Earth and causes the orbital speed of the Earth to suddenly decrease. Suggest the ways in which the orbit of the Earth will change. [2 marks]

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10f. Outline, in terms of the force acting on it, why the Earth remains in a circular orbit around the Sun. [2 marks]

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11a. Outline what is meant by escape speed. [1 mark]

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11b. A probe is launched vertically upwards from the surface of a planet with a speed [3 marks]

$$v = \frac{3}{4}v_{\text{esc}}$$

where v_{esc} is the escape speed from the planet. The planet has no atmosphere. Determine, in terms of the radius of the planet R , the maximum height from the surface of the planet reached by the probe.

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11c. The total energy of a probe in orbit around a planet of mass M is *[3 marks]*
 $E = -\frac{GMm}{2r}$ where m is the mass of the probe and r is the orbit radius.
A probe in low orbit experiences a small frictional force. Suggest the effect of this force on the speed of the probe.